# Session 1 — Arithmetic sequences: recognising and modelling

Number Sequences

15 Sep 2026 • B: 2 h – C: 1 h

**Objectives.**

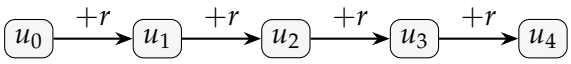
- Recognise an additive change.
- Find the common difference.
- Define a sequence in context.

**Fluency · 6 to 8 minutes, no calculator.**

1. Write the multiplier for a 10% increase. .....
2. Calculate $u_4$ if $u_0 = 14$ and $u_{n+1} = u_n + 3$. .....
3. How many terms from $u_3$ to $u_{15}$? .....
4. Give the multiplier for a 8% decrease. .....
5. Give the common difference of $12, 12+4, 12+2\times 4, \ldots$. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise an additive change and find the common difference.

**Activity 1.** A club has 320 members. It expects to gain 18 more each year. Let $u_n$ be the number of members after $n$ years. Calculate $u_0, u_1, u_2$. Which operation takes you from one term to the next? Write a relation between $u_{n+1}$ and $u_n$, then predict $u_5$.




**Written record.** A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a real-life situation, a constant absolute increase leads to an arithmetic sequence.

**Example 2.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 1.1 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 110$ and $u_{n+1} = u_n + 3$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 420 and decreases by 3% per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Application.** A warehouse manager is testing three models. For every natural number $n$, $n$ is the week number.

- Number of parcels prepared: $a_n = 3(2n + 7) - n$.
- Number of customer complaints: $b_n = 2n^2 - n + 4$.
- Number of pallets still in stock: $c_n = 150 - 7n$.

a. For each of the three sequences, calculate $u_{n+1} - u_n$ (with $u = a$, $b$ or $c$) and state whether the sequence is arithmetic; if so, give its common difference.

b. For each arithmetic sequence, interpret the common difference in context.

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 125 and the rate or common difference/ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

**Session 2** — **Arithmetic sequences: the general term**

Number Sequences — 17 Sep 2026 • B: 1 h – C: 2 h

**Objectives.**

- Move from the recurrence relation to the general term.
- Calculate a distant term.
- Interpret the index.

**Fluency · 6 to 8 minutes, no calculator.**

1. Simplify $3x + 5 - x + 2$. ……
2. How many terms from $u_3$ to $u_{15}$? ……
3. Give the multiplier for a 8% decrease. ……
4. Give the common difference of $12, 12 + 5, 12 + 2 \times 5, \ldots$. ……
5. Calculate $u_4$ if $u_0 = 15$ and $u_{n+1} = u_n + 3$. ……

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: move from the recurrence relation to the general term and calculate a distant term.

**Activity 2.** A shop has 1,250 active loyalty cards at index 0 and gains 75 per month. Without calculating all the terms, find the number of cards at index $n$, then at index 18. Explain the role of 18 in the calculation.

**Written record.** If $(u_n)$ is arithmetic with common difference $r$, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, depending on the index of the first term.

**Example 3.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 1.2 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 120$ and $u_{n+1} = u_n + 4$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 440 and decreases by 4% per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Application.** A delivery company is gradually replacing its diesel vehicles with electric ones. At the start of 2025 (rank 0), it owns 186 diesel vehicles and it withdraws 4 of them every year. Let $d_n$ be the number of diesel vehicles at the start of the year $2025 + n$; the sequence $(d_n)$ is arithmetic.

a. Give the first term $d_0$ and the common difference of the sequence $(d_n)$.

b. Express $d_n$ in terms of $n$ for every natural number $n$.

c. Calculate $d_{12}$ and $d_{30}$. Interpret $d_{30}$ in a sentence.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 130 and the rate or common difference/ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

**Session 3** — **Sum of the terms of an arithmetic sequence**

Number Sequences — 22 Sep 2026 • B: 2 h – C: 1 h

**Objectives.**

- Count the terms of a sum.
- Apply the formula.
- Check the order of magnitude.

**Fluency · 6 to 8 minutes, no calculator.**

1. Calculate 20% of 120. .....
2. Give the multiplier for a 8% decrease. .....
3. Give the common difference of $12, 12+6, 12+2\times 6, \ldots$. .....
4. Calculate $u_4$ if $u_0 = 16$ and $u_{n+1} = u_n + 3$. .....
5. How many terms from $u_3$ to $u_{15}$? .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: count the terms of a sum and apply the formula.

**Activity 3.** A company pays a monthly bonus: 45 euros in the first month, then 5 euros more each month. Calculate the total paid over the first twelve months in three ways: explicit sum, pairing first and last terms, formula.

**Written record.** A sum of arithmetic terms equals number of terms $\times \dfrac{\text{first} + \text{last}}{2}$. From $u_p$ to $u_q$, there are $q - p + 1$ terms.

**Example 4.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 1.3 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 130$ and $u_{n+1} = u_n + 5$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 460 and decreases by 5% per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Application.**

a. Calculate $\sum_{i=1}^{365} i = 1 + 2 + \cdots + 365$.

b. Calculate $\sum_{k=100}^{365} k = 100 + 101 + \cdots + 365$. State the number of terms in this sum.

c. To build up some savings, Lova takes on a challenge: on the $k$-th day of the year, she puts $k$ cents into a money box (1 cent on day 1, 2 cents on day 2, and so on). How much, in euros, will she have put in altogether over a 365-day year? And from day 100 to day 365 inclusive?

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 135 and the rate or common difference/ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

## Session 4 — Geometric sequences: recognising and modelling

Number Sequences

24 Sep 2026 • B: 1 h – C: 2 h

**Objectives.**

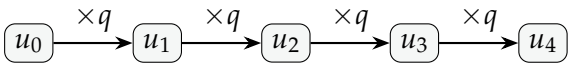
- Recognise a multiplicative change.
- Convert a rate into a multiplier.
- Compare arithmetic and geometric sequences.

**Fluency · 6 to 8 minutes, no calculator.**

1. Write the multiplier for a 25% increase. .....
2. Give the common difference of $12, 12+7, 12+2\times 7, \ldots$. .....
3. Calculate $u_4$ if $u_0 = 17$ and $u_{n+1} = u_n + 3$. .....
4. How many terms from $u_3$ to $u_{15}$? .....
5. Give the multiplier for a 8% decrease. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise a multiplicative change and convert a rate into a multiplier.

**Activity 4.** An online shop increases its number of visitors by 6% per month, starting from 8,000 visitors. Calculate the first three values. Compare the differences, then the quotients. Identify the type of sequence.




**Written record.** A sequence is geometric when $\dfrac{u_{n+1}}{u_n} = q$ is constant (non-zero terms). A constant relative change leads to a geometric sequence.

**Example 5.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 1.4 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 140$ and $u_{n+1} = u_n + 6$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 480 and decreases by 6% per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Application.** An app is being tested before its launch. In week 0, 3 testers install it. We assume that, each week, the number of new testers is multiplied by 4. Let $v_n$ be the number of new testers in week $n$; the sequence $(v_n)$ is geometric.

a. Give the first term and the common ratio of the sequence $(v_n)$, then express $v_n$ in terms of $n$.

b. Calculate $v_5$ and $v_9$.

c. Does the model seem realistic over a long period? Justify in a sentence.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 140 and the rate or common difference/ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

## Session 5 — Geometric sequences: general term and sum

Number Sequences

29 Sep 2026 • B: 2 h – C: 1 h

**Objectives.**

- Write the general term.
- Calculate a geometric sum.
- Interpret a cumulative change.

**Fluency · 6 to 8 minutes, no calculator.**

1. Simplify $3x + 8 - x + 2$. .....
2. Calculate $u_4$ if $u_0 = 18$ and $u_{n+1} = u_n + 3$. .....
3. How many terms from $u_3$ to $u_{15}$? .....
4. Give the multiplier for a 8% decrease. .....
5. Give the common difference of $12, 12 + 8, 12 + 2 \times 8, \ldots$. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: write the general term and calculate a geometric sum.

**Activity 5.** A capital of 250,000 Ar is invested at 4% per year, compound interest. Write the capital $C_n$ after $n$ years. Calculate $C_6$, then the sum $C_0 + \cdots + C_6$, and explain what this last sum represents mathematically and why it is not the final capital.

**Written record.** If $(u_n)$ is geometric with common ratio $q$, $u_n = u_0 q^n$ ou $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

**Example 6.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 1.5 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 150$ and $u_{n+1} = u_n + 2$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 500 and decreases by 2% per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Application.** A reforestation association plants 1,500 trees in the first year of its programme. Thanks to new volunteers, the number of trees planted doubles every year. Let $u_0 = 1{,}500$ be the number of trees planted in the first year and $u_n$ the number of trees planted in year $(n+1)$.

a. Justify that $(u_n)$ is a geometric sequence. Express $u_n$ in terms of $n$ and deduce $u_6$.

b. Calculate $S = u_0 + u_1 + \cdots + u_6$ and interpret this result in context.

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 145 and the rate or common difference/ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

**Session 6** — **Review and formative assessment**

Number Sequences — 1 Oct 2026 • B: 1 h – C: 2 h

**Objectives.**

- Choose the right model.
- Use the formulas.
- Analyse one's own mistakes.

**Fluency · 6 to 8 minutes, no calculator.**

1. Calculate 5% of 180. .....
2. How many terms from $u_3$ to $u_{15}$? .....
3. Give the multiplier for a 8% decrease. .....
4. Give the common difference of $12, 12+9, 12+2\times 9, \ldots$. .....
5. Calculate $u_4$ if $u_0 = 19$ and $u_{n+1} = u_n + 3$. .....

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: choose the right model and use the formulas.

**Activity 6.** Sort eight situations into three categories: arithmetic, geometric, neither. For each situation, underline the word or piece of data that justifies the choice, then choose the appropriate formula.

**Written record.** Before calculating, identify: type of sequence, first index, common difference or ratio, and the term or sum asked for. The conclusion states the unit and the index.

**Example 7.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Exercise 1.6 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 160$ and $u_{n+1} = u_n + 3$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 520 and decreases by 3% per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Application.** For each statement, say whether it is true or false. Justify your answer.

1. The price of an annual travel pass goes up by €1.50 every year. This situation can be modelled by an arithmetic sequence with common difference 1.5.
2. A car loses 15% of its value every year. This situation can be modelled by a geometric

sequence with common ratio 1.15.

3. A sum is invested in a savings account with compound interest at an annual rate of 3%, with no deposits or withdrawals. The capital can be modelled by a geometric sequence with common ratio 1.03.
4. A rainwater tank holds 1,000 L. Each day it loses 5% of its contents, then 40 L are added. The volume can be modelled by a geometric sequence with common ratio 0.95.
5. The population of a town grows by 2% per year. This situation can be modelled by an arithmetic sequence with common difference 2.
6. A loan of €2,000 is granted with simple interest at an annual rate of 6%. The amount owed can be modelled by an arithmetic sequence with common difference 120.
7. The number of monthly visits to a website decreases by 300 every month. This situation can be modelled by a geometric sequence.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 150 and the rate or common difference/ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.